INFRAHUB-ARISTA-DC

DC3 network diagrams

Standalone Arista EOS datacenter: EVPN/VXLAN over an eBGP spine-leaf fabric, with firewall, internet, cloud and branch stand-ins. Every value on these pages is read from the reviewed renders in `tests/expected/` and the containerlab topology, by `scripts/dc3_diagrams.py`.

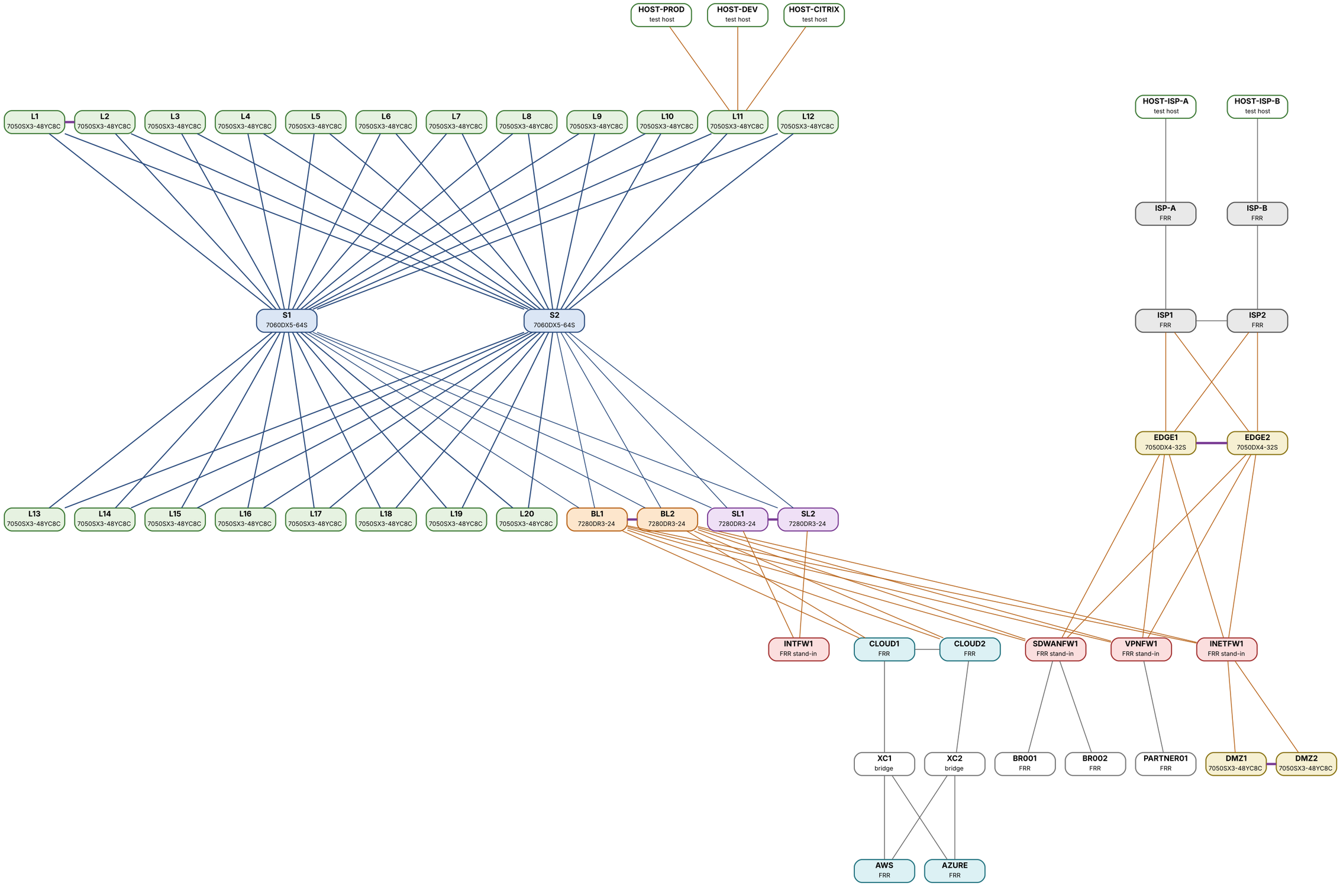
30	2	24	5	8	52	151
EOS switches	spines	VTEP leaves	MLAG pairs	stand-ins (FRR)	lab nodes	lab links

1	**Physical topology**	Every switch, stand-in and lab node, and how they are cabled
2	**Spine-leaf links**	Every uplink with its interfaces and /31 addresses (from the spine and leaf configs)
3	**Underlay eBGP**	Every VTEP leaf peers with both spines on its uplinks; MLAG pairs also peer with each other
4	**EVPN overlay**	21 VTEP addresses for 24 leaves; both spines reflect EVPN to all of them
5	**EVPN control plane: route types**	What the VTEPs tell each other over BGP EVPN, with DC3's own values
6	**Data plane: packet walks inside the fabric**	How a frame crosses the fabric when it is bridged and when it is routed
7	**Data plane: packet walk out through the firewall (captured)**	PROD host to 198.18.1.10: the fabric is crossed twice, and the VRF changes at the Internal FW
8	**MLAG pairs**	Five pairs: peer-link members, peer VLAN, keepalive, VTEP loopbacks and the MLAG port-channels to the stand-ins
9	**Service leaves and Internal FW**	Each tenant VRF and TRANSIT meet the Internal FW on its own VLAN; eBGP per VRF, BFD 999 ms x 3
10	**Border leaves and the outside**	TRANSIT (and PARTNER) leave the fabric here: Internet FW, SD-WAN hub, VPN partner FW, two cloud routers
11	**Internet edge, DMZ, clouds and branches**	Everything outside the VTEP fabric, with interfaces and addresses from the stand-in scripts
12	**VRFs and traffic paths**	Where each VRF lives and the three ways traffic leaves the fabric
13	**Routing policy**	Every route-map on the firewall, cloud and SD-WAN attachments (service pair and border pair; both members are equal)
14	**Tenant segments**	42 server segments on every server leaf (L1..L20), same VLAN, VNI and gateway everywhere
15	**Management and ZTP**	OOB addresses from pool DC3-MGMT0 (10.250.32.0/24, gateway .1) and how a blank switch gets its config
16	**Lab and tests**	How the lab is run, and what each failure test breaks

Source commit 483d87b · generated 2026-10-10 01:07 UTC · design: docs/DESIGN.md · lab: docs/QUICKSTART.md

1 · Physical topology

Every switch, stand-in and lab node, and how they are cabled



Spine (EOS) · Server leaf (EOS) · Border leaf (EOS) · Service leaf (EOS) · Edge / DMZ pair (EOS, L2) · Firewall stand-in (FRR) · ISP router / carrier (FRR) · Cloud router / provider (FRR) · Branch, partner, cross-connect (lab only) · Test host (lab only)

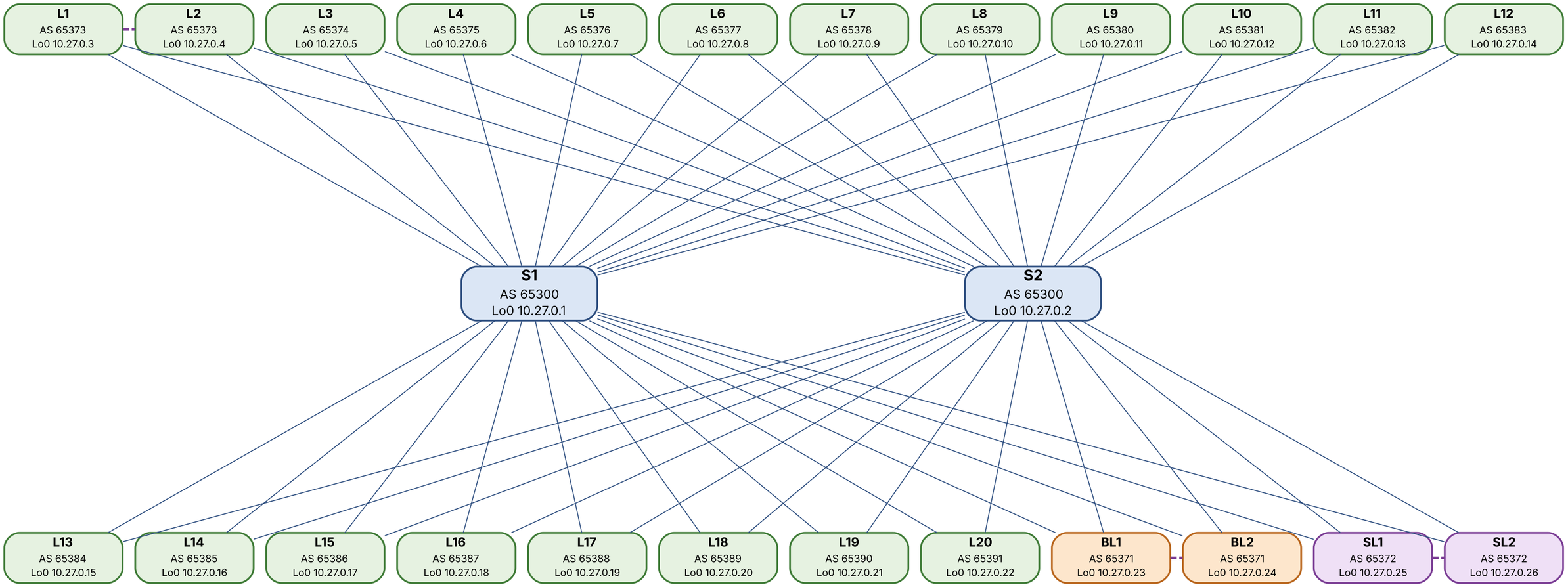
52 nodes, 151 links (lab topology). Blue: spine-leaf links (two per spine on server leaves, one on border and service leaves; page 2). Purple: MLAG peer-links (keepalives on page 8). Orange: switch to stand-in attachments (MLAG port-channels, pages 8-11). Grey: links outside the switches (lab only, page 11). Models are the design's hardware (docs/DESIGN.md section 2); the lab runs cEOS-lab.

2 · Spine-leaf links

Every uplink with its interfaces and /31 addresses (from the spine and leaf configs)

Leaf	ASN	Uplink 1	Uplink 2	Uplink 3	Uplink 4
BL1	65371	Et1/1 10.27.4.81/31 ↔ S1 Et41/1 10.27.4.80/31	Et2/1 10.27.4.169/31 ↔ S2 Et41/1 10.27.4.168/31		
BL2	65371	Et1/1 10.27.4.83/31 ↔ S1 Et42/1 10.27.4.82/31	Et2/1 10.27.4.171/31 ↔ S2 Et42/1 10.27.4.170/31		
SL1	65372	Et1/1 10.27.4.85/31 ↔ S1 Et43/1 10.27.4.84/31	Et2/1 10.27.4.173/31 ↔ S2 Et43/1 10.27.4.172/31		
SL2	65372	Et1/1 10.27.4.87/31 ↔ S1 Et44/1 10.27.4.86/31	Et2/1 10.27.4.175/31 ↔ S2 Et44/1 10.27.4.174/31		
L1	65373	Et49/1 10.27.4.1/31 ↔ S1 Et1/1 10.27.4.0/31	Et50/1 10.27.4.3/31 ↔ S1 Et2/1 10.27.4.2/31	Et51/1 10.27.4.89/31 ↔ S2 Et1/1 10.27.4.88/31	Et52/1 10.27.4.91/31 ↔ S2 Et2/1 10.27.4.90/31
L2	65373	Et49/1 10.27.4.5/31 ↔ S1 Et3/1 10.27.4.4/31	Et50/1 10.27.4.7/31 ↔ S1 Et4/1 10.27.4.6/31	Et51/1 10.27.4.93/31 ↔ S2 Et3/1 10.27.4.92/31	Et52/1 10.27.4.95/31 ↔ S2 Et4/1 10.27.4.94/31
L3	65374	Et49/1 10.27.4.9/31 ↔ S1 Et5/1 10.27.4.8/31	Et50/1 10.27.4.11/31 ↔ S1 Et6/1 10.27.4.10/31	Et51/1 10.27.4.97/31 ↔ S2 Et5/1 10.27.4.96/31	Et52/1 10.27.4.99/31 ↔ S2 Et6/1 10.27.4.98/31
L4	65375	Et49/1 10.27.4.13/31 ↔ S1 Et7/1 10.27.4.12/31	Et50/1 10.27.4.15/31 ↔ S1 Et8/1 10.27.4.14/31	Et51/1 10.27.4.101/31 ↔ S2 Et7/1 10.27.4.100/31	Et52/1 10.27.4.103/31 ↔ S2 Et8/1 10.27.4.102/31
L5	65376	Et49/1 10.27.4.17/31 ↔ S1 Et9/1 10.27.4.16/31	Et50/1 10.27.4.19/31 ↔ S1 Et10/1 10.27.4.18/31	Et51/1 10.27.4.105/31 ↔ S2 Et9/1 10.27.4.104/31	Et52/1 10.27.4.107/31 ↔ S2 Et10/1 10.27.4.106/31
L6	65377	Et49/1 10.27.4.21/31 ↔ S1 Et11/1 10.27.4.20/31	Et50/1 10.27.4.23/31 ↔ S1 Et12/1 10.27.4.22/31	Et51/1 10.27.4.109/31 ↔ S2 Et11/1 10.27.4.108/31	Et52/1 10.27.4.111/31 ↔ S2 Et12/1 10.27.4.110/31
L7	65378	Et49/1 10.27.4.25/31 ↔ S1 Et13/1 10.27.4.24/31	Et50/1 10.27.4.27/31 ↔ S1 Et14/1 10.27.4.26/31	Et51/1 10.27.4.113/31 ↔ S2 Et13/1 10.27.4.112/31	Et52/1 10.27.4.115/31 ↔ S2 Et14/1 10.27.4.114/31
L8	65379	Et49/1 10.27.4.29/31 ↔ S1 Et15/1 10.27.4.28/31	Et50/1 10.27.4.31/31 ↔ S1 Et16/1 10.27.4.30/31	Et51/1 10.27.4.117/31 ↔ S2 Et15/1 10.27.4.116/31	Et52/1 10.27.4.119/31 ↔ S2 Et16/1 10.27.4.118/31
L9	65380	Et49/1 10.27.4.33/31 ↔ S1 Et17/1 10.27.4.32/31	Et50/1 10.27.4.35/31 ↔ S1 Et18/1 10.27.4.34/31	Et51/1 10.27.4.121/31 ↔ S2 Et17/1 10.27.4.120/31	Et52/1 10.27.4.123/31 ↔ S2 Et18/1 10.27.4.122/31
L10	65381	Et49/1 10.27.4.37/31 ↔ S1 Et19/1 10.27.4.36/31	Et50/1 10.27.4.39/31 ↔ S1 Et20/1 10.27.4.38/31	Et51/1 10.27.4.125/31 ↔ S2 Et19/1 10.27.4.124/31	Et52/1 10.27.4.127/31 ↔ S2 Et20/1 10.27.4.126/31
L11	65382	Et49/1 10.27.4.41/31 ↔ S1 Et21/1 10.27.4.40/31	Et50/1 10.27.4.43/31 ↔ S1 Et22/1 10.27.4.42/31	Et51/1 10.27.4.129/31 ↔ S2 Et21/1 10.27.4.128/31	Et52/1 10.27.4.131/31 ↔ S2 Et22/1 10.27.4.130/31
L12	65383	Et49/1 10.27.4.45/31 ↔ S1 Et23/1 10.27.4.44/31	Et50/1 10.27.4.47/31 ↔ S1 Et24/1 10.27.4.46/31	Et51/1 10.27.4.133/31 ↔ S2 Et23/1 10.27.4.132/31	Et52/1 10.27.4.135/31 ↔ S2 Et24/1 10.27.4.134/31
L13	65384	Et49/1 10.27.4.49/31 ↔ S1 Et25/1 10.27.4.48/31	Et50/1 10.27.4.51/31 ↔ S1 Et26/1 10.27.4.50/31	Et51/1 10.27.4.137/31 ↔ S2 Et25/1 10.27.4.136/31	Et52/1 10.27.4.139/31 ↔ S2 Et26/1 10.27.4.138/31
L14	65385	Et49/1 10.27.4.53/31 ↔ S1 Et27/1 10.27.4.52/31	Et50/1 10.27.4.55/31 ↔ S1 Et28/1 10.27.4.54/31	Et51/1 10.27.4.141/31 ↔ S2 Et27/1 10.27.4.140/31	Et52/1 10.27.4.143/31 ↔ S2 Et28/1 10.27.4.142/31
L15	65386	Et49/1 10.27.4.57/31 ↔ S1 Et29/1 10.27.4.56/31	Et50/1 10.27.4.59/31 ↔ S1 Et30/1 10.27.4.58/31	Et51/1 10.27.4.145/31 ↔ S2 Et29/1 10.27.4.144/31	Et52/1 10.27.4.147/31 ↔ S2 Et30/1 10.27.4.146/31
L16	65387	Et49/1 10.27.4.61/31 ↔ S1 Et31/1 10.27.4.60/31	Et50/1 10.27.4.63/31 ↔ S1 Et32/1 10.27.4.62/31	Et51/1 10.27.4.149/31 ↔ S2 Et31/1 10.27.4.148/31	Et52/1 10.27.4.151/31 ↔ S2 Et32/1 10.27.4.150/31
L17	65388	Et49/1 10.27.4.65/31 ↔ S1 Et33/1 10.27.4.64/31	Et50/1 10.27.4.67/31 ↔ S1 Et34/1 10.27.4.66/31	Et51/1 10.27.4.153/31 ↔ S2 Et33/1 10.27.4.152/31	Et52/1 10.27.4.155/31 ↔ S2 Et34/1 10.27.4.154/31
L18	65389	Et49/1 10.27.4.69/31 ↔ S1 Et35/1 10.27.4.68/31	Et50/1 10.27.4.71/31 ↔ S1 Et36/1 10.27.4.70/31	Et51/1 10.27.4.157/31 ↔ S2 Et35/1 10.27.4.156/31	Et52/1 10.27.4.159/31 ↔ S2 Et36/1 10.27.4.158/31
L19	65390	Et49/1 10.27.4.73/31 ↔ S1 Et37/1 10.27.4.72/31	Et50/1 10.27.4.75/31 ↔ S1 Et38/1 10.27.4.74/31	Et51/1 10.27.4.161/31 ↔ S2 Et37/1 10.27.4.160/31	Et52/1 10.27.4.163/31 ↔ S2 Et38/1 10.27.4.162/31
L20	65391	Et49/1 10.27.4.77/31 ↔ S1 Et39/1 10.27.4.76/31	Et50/1 10.27.4.79/31 ↔ S1 Et40/1 10.27.4.78/31	Et51/1 10.27.4.165/31 ↔ S2 Et39/1 10.27.4.164/31	Et52/1 10.27.4.167/31 ↔ S2 Et40/1 10.27.4.166/31

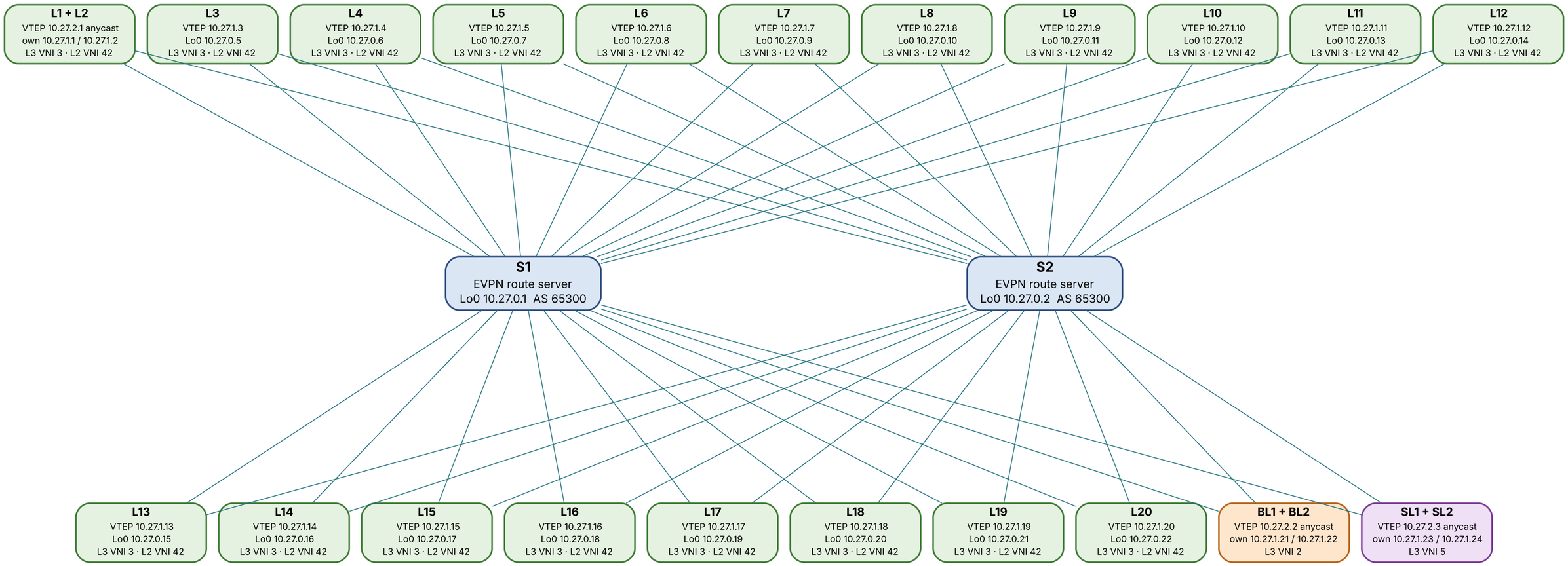
88 spine-leaf links, each a routed /31 from pool DC3-FABRIC-P2P (10.27.4.0/22), MTU 9214, BFD 999 ms x 3, eBGP in peer group IPV4-UNDERLAY-PEERS. Leaf side first; addresses as configured.



Switch	ASN	Router ID (Lo0)	Lo1 (VTEP)	Lo2 (own VTEP)	Underlay sessions	MLAG iBGP peer
BL1	65371	10.27.0.23	10.27.2.2	10.27.1.21	2	10.27.3.3
BL2	65371	10.27.0.24	10.27.2.2	10.27.1.22	2	10.27.3.2
L1	65373	10.27.0.3	10.27.2.1	10.27.1.1	4	10.27.3.1
L2	65373	10.27.0.4	10.27.2.1	10.27.1.2	4	10.27.3.0
L3	65374	10.27.0.5	10.27.1.3		4	
L4	65375	10.27.0.6	10.27.1.4		4	
L5	65376	10.27.0.7	10.27.1.5		4	
L6	65377	10.27.0.8	10.27.1.6		4	
L7	65378	10.27.0.9	10.27.1.7		4	
L8	65379	10.27.0.10	10.27.1.8		4	
L9	65380	10.27.0.11	10.27.1.9		4	
L10	65381	10.27.0.12	10.27.1.10		4	
L11	65382	10.27.0.13	10.27.1.11		4	

Switch	ASN	Router ID (Lo0)	Lo1 (VTEP)	Lo2 (own VTEP)	Underlay sessions	MLAG iBGP peer
L12	65383	10.27.0.14	10.27.1.12		4	
L13	65384	10.27.0.15	10.27.1.13		4	
L14	65385	10.27.0.16	10.27.1.14		4	
L15	65386	10.27.0.17	10.27.1.15		4	
L16	65387	10.27.0.18	10.27.1.16		4	
L17	65388	10.27.0.19	10.27.1.17		4	
L18	65389	10.27.0.20	10.27.1.18		4	
L19	65390	10.27.0.21	10.27.1.19		4	
L20	65391	10.27.0.22	10.27.1.20		4	
S1	65300	10.27.0.1			44	
S2	65300	10.27.0.2			44	
SL1	65372	10.27.0.25	10.27.2.3	10.27.1.23	2	10.27.3.5
SL2	65372	10.27.0.26	10.27.2.3	10.27.1.24	2	10.27.3.4

One ASN per leaf or MLAG pair, spines AS 65300. Loopbacks are announced with redistribute connected route-map RM-CONN-2-BGP (PL-LOOPBACKS, eq 32). ECMP maximum-paths 4. Blue: one line per spine and leaf (2 links each on server leaves, page 2). Purple dashed: iBGP between MLAG members over Vlan3900. EDGE and DMZ pairs run no BGP.

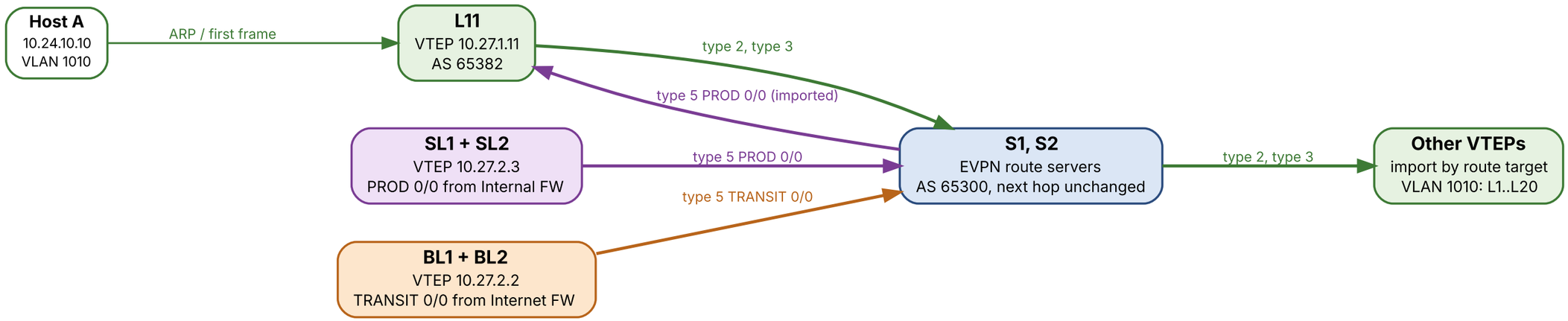


VRF	L3 VNI	Route target	Switches	Where
PROD	50001	65300:50001	22	L1–L20, SL1, SL2
DEV	50002	65300:50002	22	L1–L20, SL1, SL2
CITRIX	50022	65300:50022	22	L1–L20, SL1, SL2
PARTNER	50031	65300:50031	4	BL1, BL2, SL1, SL2
TRANSIT	50100	65300:50100	4	BL1, BL2, SL1, SL2

EVPN over eBGP multihop (TTL 3) from each switch's Loopback0 to both spines' Loopback0, send-community, multihop BFD 999 ms x 3. RD = <Loopback0>:<VNI>, RT = 65300:<VNI>. MLAG pairs share one anycast VTEP on Loopback1 (vxlan virtual-router encapsulation mac-address mlag-system-id). VXLAN UDP 4789. Anycast gateway MAC 0200.0000.00aa.

5 · EVPN control plane: route types

What the VTEPs tell each other over BGP EVPN, with DC3's own values



Type 2 · MAC/IP advertisement — captured

Originated by	L11 (AS 65382) when it learns host A (redistribute learned, VLAN 1010)
RD	10.27.0.13:11010 (Loopback0 : L2 VNI)
ESI / Ethernet tag	0 / 0 (single-homed host)
MAC	aa:c1:ab:09:17:8d (host A)
IP	10.24.10.10 (a second route carries the MAC only)
Label 1 / Label 2	11010 (L2 VNI, bridging) / 50001 (L3 VNI of PROD, routing)
Next hop	10.27.1.11 (L11 VTEP, Loopback1); the spines keep it
Route targets	65300:11010, 65300:50001
Ext. communities	VXLAN encapsulation · router MAC `00:1c:73:6f:ff:35`
AS path at a receiver	65300 65382
Used for	bridging to host A in VLAN 1010, and a /32 host route in PROD

Type 3 · Inclusive multicast (IMET) — flood list (show vxlan vtep: 18)

Originated by	every VTEP for each of its L2 VNIs (L1..L20: 42 each; BL/SL: none)
RD	10.27.0.13:11010
Ethernet tag	0
Originating IP	10.27.1.11
PMSI tunnel	ingress replication, label 11010, endpoint 10.27.1.11 (RFC 8365 form; no type-3 update was in the capture)
Route target	65300:11010
Builds	the flood list of VNI 11010: L11 sends BUM copies to 18 VTEPs
Flood list on L11	10.27.1.3, 10.27.1.4, 10.27.1.5, 10.27.1.6, 10.27.1.7, 10.27.1.8, 10.27.1.9, 10.27.1.10, 10.27.1.12, 10.27.1.13, 10.27.1.14, 10.27.1.15, 10.27.1.16, 10.27.1.17, 10.27.1.18, 10.27.1.19, 10.27.1.20, 10.27.2.1

Type 5 · IP prefix — captured

Originated by	SL pair, from the Internal FW's 0/0 in VRF PROD
RD	10.27.0.25:50001
Prefix	0.0.0.0/0 (gateway IP 0)
Label	50001 (L3 VNI of PROD)
Next hop	10.27.2.3 (SL anycast VTEP)
Route target	65300:50001
Ext. communities	VXLAN · router MAC `02:1c:73:5f:aa:c6`
Installed on L11 as	`0.0.0.0/0 via VTEP 10.27.2.3 VNI 50001 router-mac 02:1c:73:5f:aa:c6`
Not imported	the BL pair's TRANSIT 0/0 (RD 10.27.0.23:50100, RT 65300:50100, next hop 10.27.2.2): L11 has no TRANSIT VRF

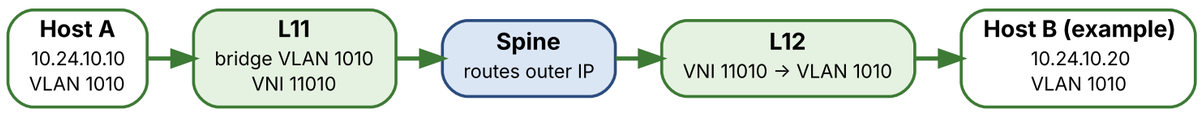
Switches	Type 2 (MAC/IP)	Type 3 (IMET)	Type 5 (IP prefix)
Server leaves L1..L20	hosts learned in their 42 VLANs (redistribute learned)	one per L2 VNI (42)	connected SVI subnets per VRF (redistribute connected)
Service leaves SL1, SL2	–	–	routes from the Internal FW: 0/0 into each tenant VRF, tenant summaries into TRANSIT; aggregates
Border leaves BL1, BL2	–	–	TRANSIT: 0/0, cloud and branch prefixes from the outside; PARTNER: partner NAT range
Spines S1, S2	pass on, add AS 65300	pass on	pass on (not VTEPs, next hop unchanged)

RD and RT rules: RD = <Loopback0>:<VNI>, RT = 65300:<VNI>. Values marked captured were seen in the full lab (host and router MACs are lab container values); the rest is read from the configs. Type 1 and type 4 (Ethernet segment routes) come with the ESI server ports, which are deferred.

6 · Data plane: packet walks inside the fabric

How a frame crosses the fabric when it is bridged and when it is routed

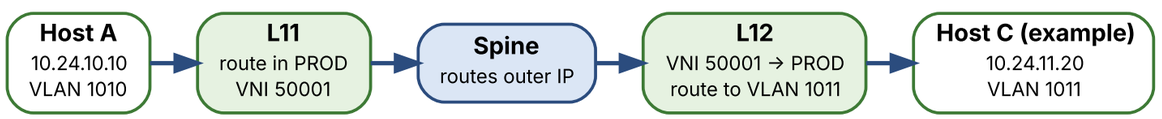
Walk 1 · Bridged, same VLAN on two leaves (L2 VNI 11010) derived from the configs not captured: the lab hosts all sit on L11



1. A sends ARP for B (broadcast in VLAN 1010). If L11 has no type-2 route for B yet, it copies the frame to every VTEP in the VNI 11010 flood list (18 VTEPs, built from type-3 routes) — ingress replication, one unicast VXLAN copy each.
2. L12 decapsulates, floods into VLAN 1010; B answers. L12 already has A's type-2 route, so the reply goes straight to 10.27.1.11.
3. Both leaves now advertise type-2 routes for A and B. Every next frame is unicast VXLAN with VNI 11010: the leaves only bridge, no routing, inner MACs are the hosts' own.

Outer Ethernet 14 B	**Outer IPv4** 20 B	**UDP** 8 B	**VXLAN** 8 B	**Inner Ethernet** 14 B	**Inner IPv4** 20 B	**Payload** 64 B
L11 uplink → spine	10.27.1.11 → 10.27.1.12 UDP payload	src: flow hash dst 4789	VNI 11010 L2 VNI of VLAN 1010	A MAC → B MAC VLAN tag removed	10.24.10.10 → 10.24.10.20 TTL unchanged	ICMP echo

Walk 2 · Routed, two subnets of one VRF on two leaves (symmetric IRB, L3 VNI 50001) derived from the configs not captured



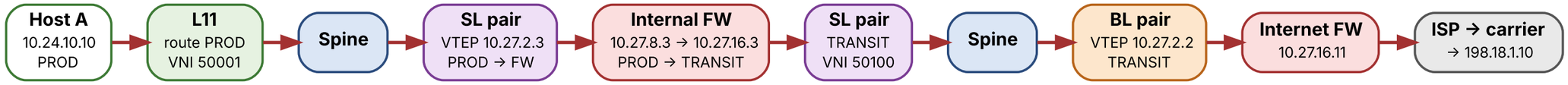
1. A sends to its gateway 10.24.10.1, destination MAC 02:00:00:00:00:aa (the same anycast gateway on every leaf).
2. L11 routes in VRF PROD. C's type-2 MAC/IP route (label 2 = 50001, router MAC of L12) gives a /32 host route via VTEP 10.27.1.12. Without it, the connected-subnet type-5 route of 10.24.11.0/24 is used, and the leaf that receives the packet ARPs for C.
3. L11 encapsulates with the L3 VNI 50001 and L12's router MAC as inner destination (symmetric IRB: routed on both leaves).
4. L12 decapsulates; VNI 50001 maps to VRF PROD; it routes into VLAN 1011 and sends to C with source MAC = gateway MAC.

Outer Ethernet 14 B	**Outer IPv4** 20 B	**UDP** 8 B	**VXLAN** 8 B	**Inner Ethernet** 14 B	**Inner IPv4** 20 B	**Payload** 64 B
L11 uplink → spine	10.27.1.11 → 10.27.1.12	dst 4789	VNI 50001 L3 VNI of PROD	L11 router MAC → L12 router MAC	10.24.10.10 → 10.24.11.20 TTL −1 at L11	ICMP echo

Host B (10.24.10.20 on L12) and host C (10.24.11.20 on L12) are example hosts, not lab nodes. VTEP addresses, VNIs and the flood list are read from the configs. VXLAN adds 50 bytes; the underlay MTU is 9214, so 1500-byte host packets never fragment. The outer UDP source port is a hash of the inner flow, so the spines spread flows by ECMP.

7 · Data plane: packet walk out through the firewall (captured)

PROD host to 198.18.1.10: the fabric is crossed twice, and the VRF changes at the Internal FW



1. On the host port (L11 Ethernet1): no VXLAN yet; destination is the anycast gateway MAC.

Ethernet 14 B	**IPv4** 20 B	**ICMP** 64 B
aa:c1:ab:09:17:8d → 02:00:00:00:00:aa (gateway)	10.24.10.10 → 198.18.1.10 TTL 64	echo request

2. Leg 1, L11 → SL pair: routed in PROD, sent on the L3 VNI 50001 to the SL anycast VTEP; inner destination = SL router MAC from the type-5 0/0 (148-byte frame = 98 + 50).

Outer Ethernet 14 B	**Outer IPv4** 20 B	**UDP** 8 B	**VXLAN** 8 B	**Inner Ethernet** 14 B	**Inner IPv4** 20 B	**ICMP** 64 B
aa:c1:ab:ff:46:0d → aa:c1:ab:4c:c4:37	10.27.1.11 → 10.27.2.3 TTL 64, DF	64659 → 4789	flags I (0x08) VNI 50001	00:1c:73:6f:ff:35 (L11) → 02:1c:73:5f:aa:c6 (SL pair)	10.24.10.10 → 198.18.1.10 TTL 63	echo request

3. SL pair: decapsulates into PROD, hands the packet to the Internal FW, gets it back in TRANSIT, re-encapsulates. Leg 2, SL → BL:

Outer Ethernet 14 B	**Outer IPv4** 20 B	**UDP** 8 B	**VXLAN** 8 B	**Inner Ethernet** 14 B	**Inner IPv4** 20 B	**ICMP** 64 B
SL uplink → spine	10.27.2.3 → 10.27.2.2	→ 4789	VNI 50100 L3 VNI of TRANSIT	SL router MAC → 02:1c:73:5d:31:8a (BL pair)	10.24.10.10 → 198.18.1.10 after the FW	echo request

4. BL pair decapsulates into TRANSIT and routes to the Internet FW; from there it is plain IP to the ISP and carrier.

5. The reply, SL pair → L11 (same VNI 50001; the source is the answering SL member's own MAC, which member was not checked):

Outer Ethernet 14 B	**Outer IPv4** 20 B	**UDP** 8 B	**VXLAN** 8 B	**Inner Ethernet** 14 B	**Inner IPv4** 20 B	**ICMP** 64 B
spine → L11 uplink	10.27.2.3 → 10.27.1.11 TTL 63	59864 → 4789	flags I (0x08) VNI 50001	00:1c:73:f2:be:3c (SL member) → 00:1c:73:6f:ff:35 (L11)	198.18.1.10 → 10.24.10.10 TTL 57	echo reply

Frames counted in 40 s:

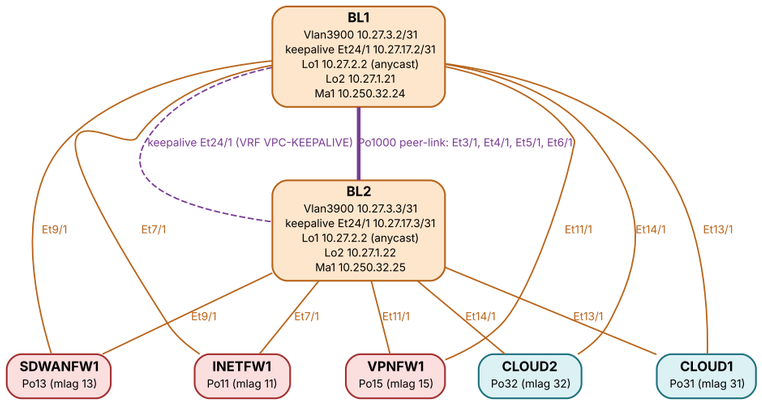
Capture point	Outer flow	VNI	Frames
L11 uplinks	10.27.1.11 → 10.27.2.3 / back	50001	3137 / 3134
SL1 uplinks	10.27.1.11 → 10.27.2.3 / back	50001	2550 / 3133
SL1 uplinks	10.27.2.3 → 10.27.2.2 / back	50100	3131 / 3130
BL1 uplinks	10.27.2.3 → 10.27.2.2 / back	50100	586 / 3132

Header values were captured in the full lab (tcpdump on L11, SL1 and BL1; docs/VXLAN-EVPN-CAPTURE.md); the outer and inner MACs of leg 2 were not decoded and are named generically. SL1 and BL1 each see part of the traffic toward their pair: both members own the anycast VTEP and the spines split flows by ECMP. Only VNIs 50001 and 50100 appear: the path is routed end to end.

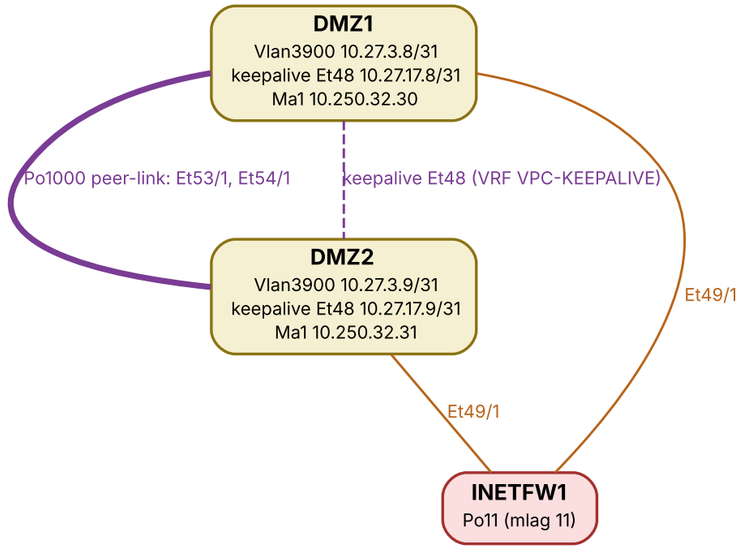
8 · MLAG pairs

Five pairs: peer-link members, peer VLAN, keepalive, VTEP loopbacks and the MLAG port-channels to the stand-ins

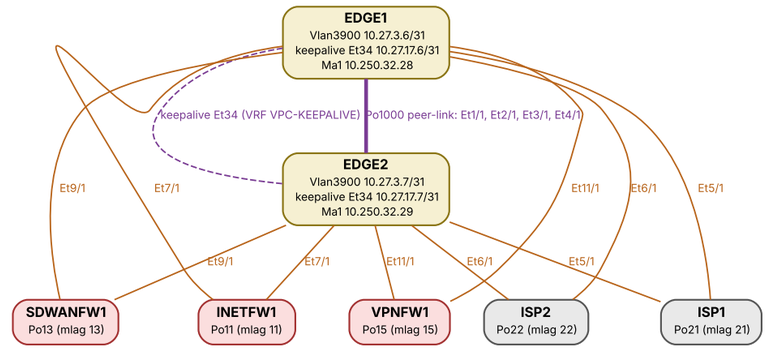
DC3-MLAG-BL1-BL2 · peer-address 10.27.3.3 / 10.27.3.2 · heartbeat 4000 ms · dual-primary delay 5 action errdisable all-interfaces · recovery mlag 360 non-mlag 300



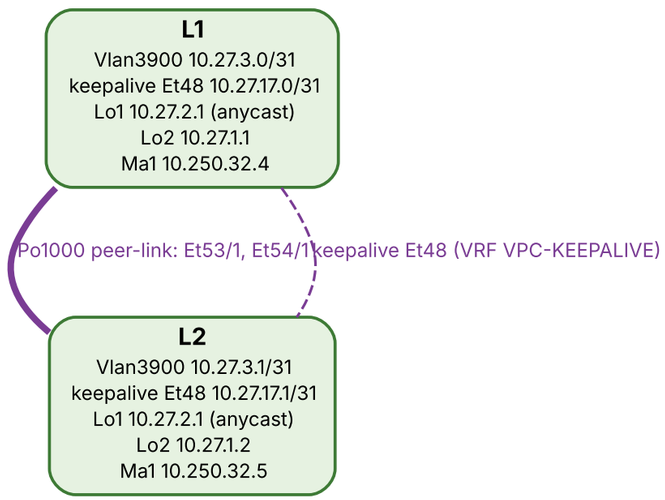
DC3-MLAG-DMZ1-DMZ2 · peer-address 10.27.3.9 / 10.27.3.8 · heartbeat 4000 ms · dual-primary delay 5 action errdisable all-interfaces · recovery mlag 360 non-mlag 300



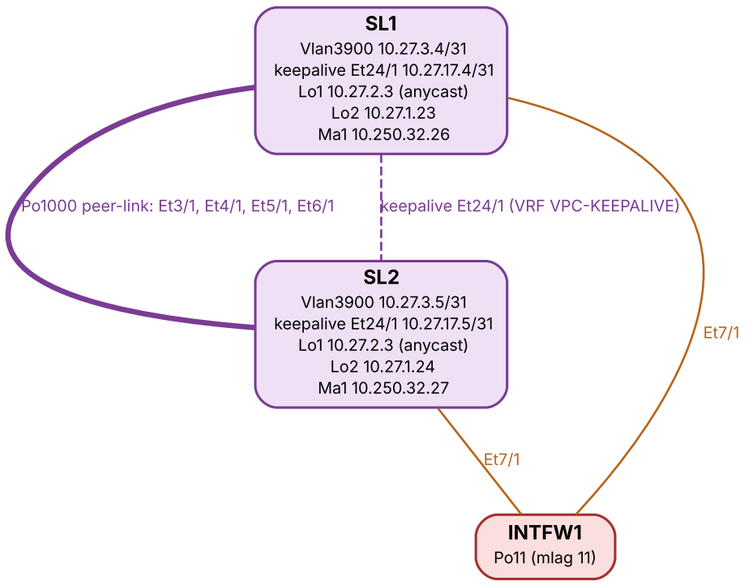
DC3-MLAG-EDGE1-EDGE2 · peer-address 10.27.3.7 / 10.27.3.6 · heartbeat 4000 ms · dual-primary delay 5 action errdisable all-interfaces · recovery mlag 360 non-mlag 300



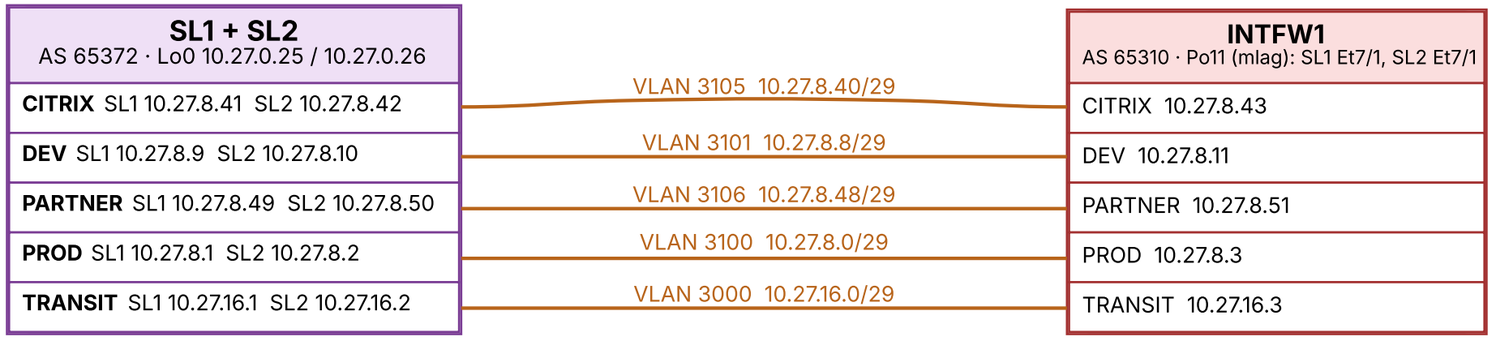
DC3-MLAG-L1-L2 · peer-address 10.27.3.1 / 10.27.3.0 · heartbeat 4000 ms · dual-primary delay 5 action errdisable all-interfaces · recovery mlag 360 non-mlag 300



DC3-MLAG-SL1-SL2 · peer-address 10.27.3.5 / 10.27.3.4 · heartbeat 4000 ms · dual-primary delay 5 action errdisable all-interfaces · recovery mlag 360 non-mlag 300

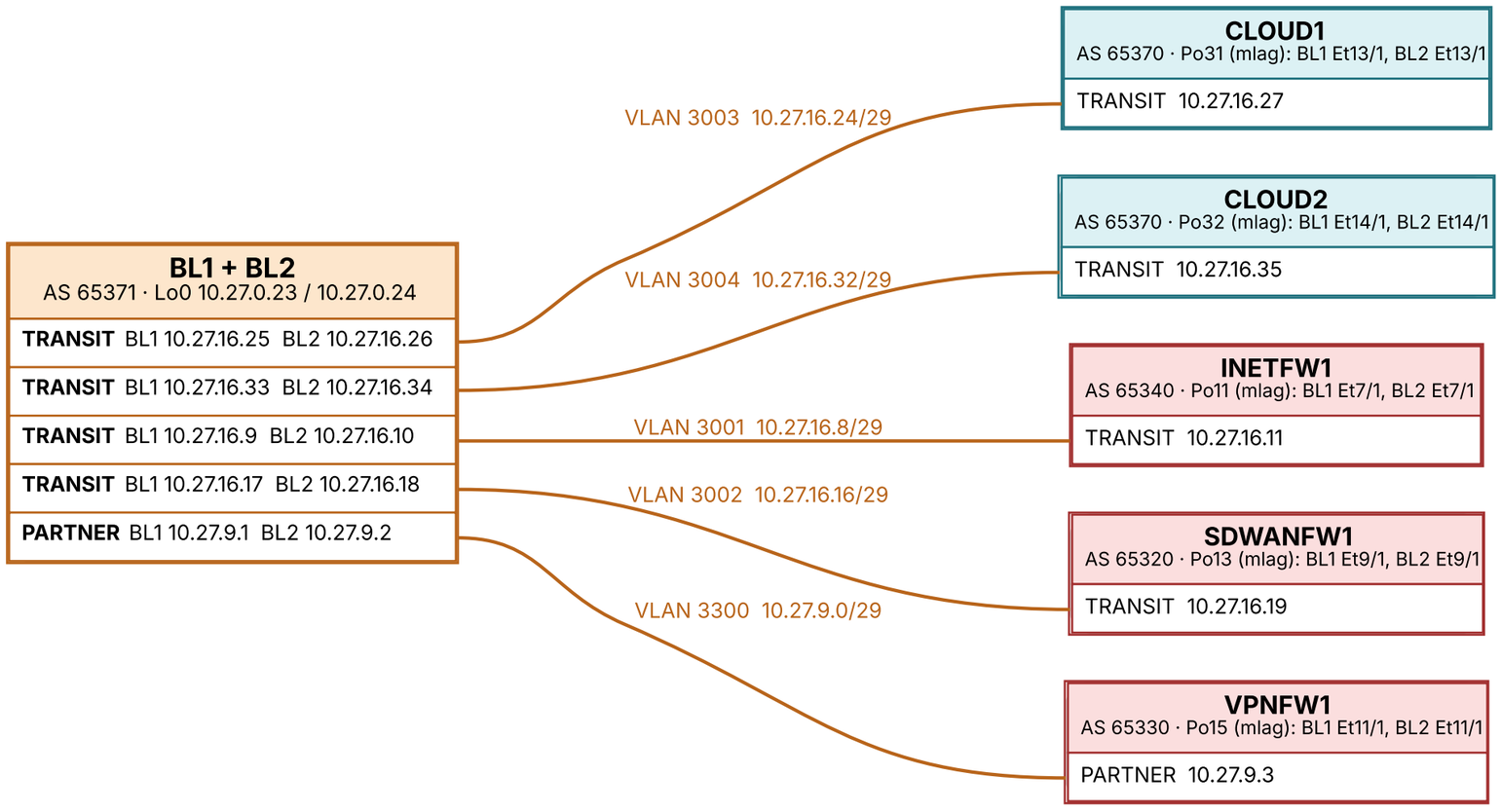


Peer-link Port-Channel1000 (LACP active, trunk group MLAGPEER), peer VLAN 3900 with no autostate, heartbeat on a dedicated routed port in VRF VPC-KEEPALIVE. mlag id = port-channel number. ESI pairs L3-L4 .. L9-L10 are EVPN multihoming pairs with no peer-link; their server Ethernet segments are deferred (docs/DESIGN.md section 11).



VRF	VLAN	Subnet	Switch SVIs	Peer	Peer AS	route-map in	route-map out
CITRIX	3105	10.27.8.40/29	SL1 10.27.8.41 / SL2 10.27.8.42	INTFW1 10.27.8.43	65310	RM-CITRIX-FROM-FW	RM-CITRIX-TO-FW
DEV	3101	10.27.8.8/29	SL1 10.27.8.9 / SL2 10.27.8.10	INTFW1 10.27.8.11	65310	RM-DEV-FROM-FW	RM-DEV-TO-FW
PARTNER	3106	10.27.8.48/29	SL1 10.27.8.49 / SL2 10.27.8.50	INTFW1 10.27.8.51	65310	RM-PARTNER-FROM-FW	RM-PARTNER-TO-FW
PROD	3100	10.27.8.0/29	SL1 10.27.8.1 / SL2 10.27.8.2	INTFW1 10.27.8.3	65310	RM-PROD-FROM-FW	RM-PROD-TO-FW
TRANSIT	3000	10.27.16.0/29	SL1 10.27.16.1 / SL2 10.27.16.2	INTFW1 10.27.16.3	65310	RM-TRANSIT-FROM-FW	RM-TRANSIT-TO-FW

SVI .1 / .2 on the pair members, the stand-in on .3 of each /29, all over one MLAG port-channel (Po11). Tenant VRFs send their DC3 summary and accept only 0/0; in TRANSIT the service leaves send only 0/0 and accept the tenant summaries. Page 13 lists the prefix-lists.



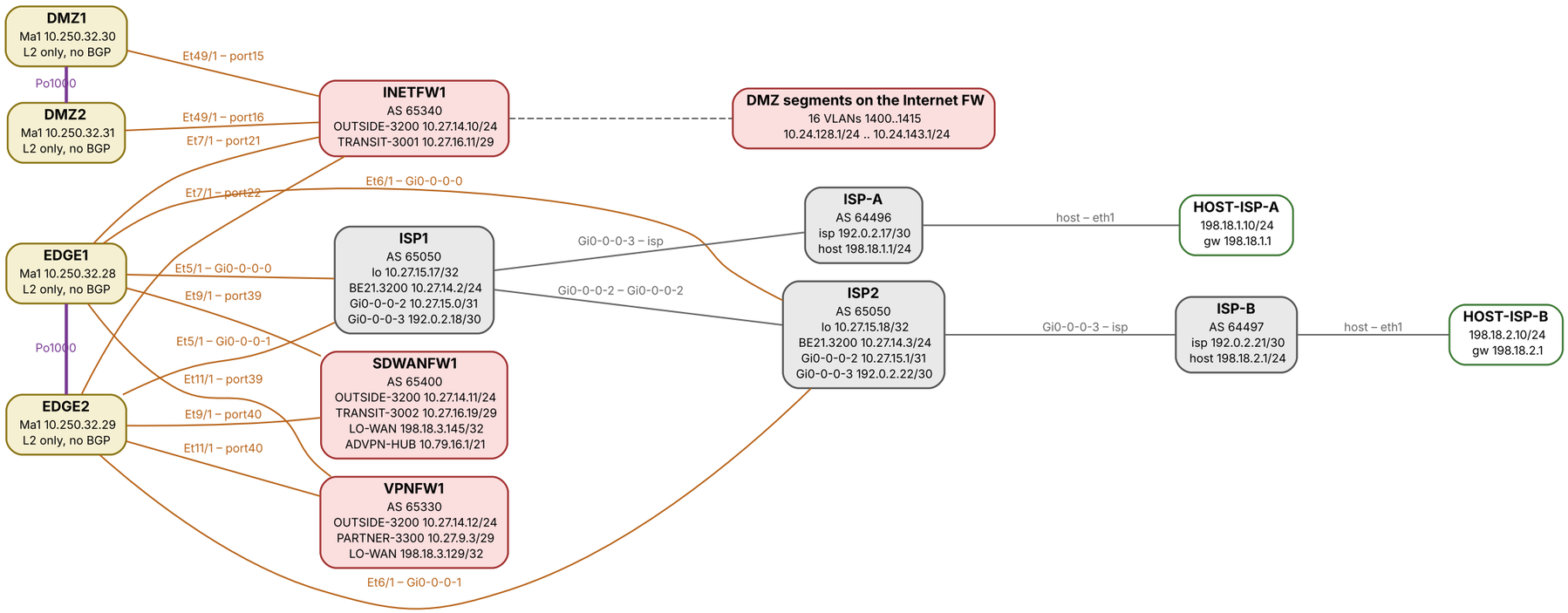
VRF	VLAN	Subnet	Switch SVIs	Peer	Peer AS	route-map in	route-map out
TRANSIT	3003	10.27.16.24/29	BL1 10.27.16.25 / BL2 10.27.16.26	CLOUD1 10.27.16.27	65370	RM-TRANSIT-FROM-CLOUD	RM-TRANSIT-TO-CLOUD
TRANSIT	3004	10.27.16.32/29	BL1 10.27.16.33 / BL2 10.27.16.34	CLOUD2 10.27.16.35	65370	RM-TRANSIT-FROM-CLOUD	RM-TRANSIT-TO-CLOUD
TRANSIT	3001	10.27.16.8/29	BL1 10.27.16.9 / BL2 10.27.16.10	INETFW1 10.27.16.11	65340	RM-TRANSIT-FROM-FW	RM-TRANSIT-TO-FW
TRANSIT	3002	10.27.16.16/29	BL1 10.27.16.17 / BL2 10.27.16.18	SDWANFW1 10.27.16.19	65320	RM-TRANSIT-FROM-SDWAN	RM-TRANSIT-TO-SDWAN
PARTNER	3300	10.27.9.0/29	BL1 10.27.9.1 / BL2 10.27.9.2	VPNFW1 10.27.9.3	65330	RM-PARTNER-FROM-FW	RM-PARTNER-TO-FW

The border leaves carry TRANSIT to the Internet FW (accept 0/0 only, send the tenant summaries), to the SD-WAN hub (branch summary), to both cloud routers (cloud prefixes) and PARTNER to the VPN FW. One MLAG port-channel per stand-in (Po11/13/15/31/32).

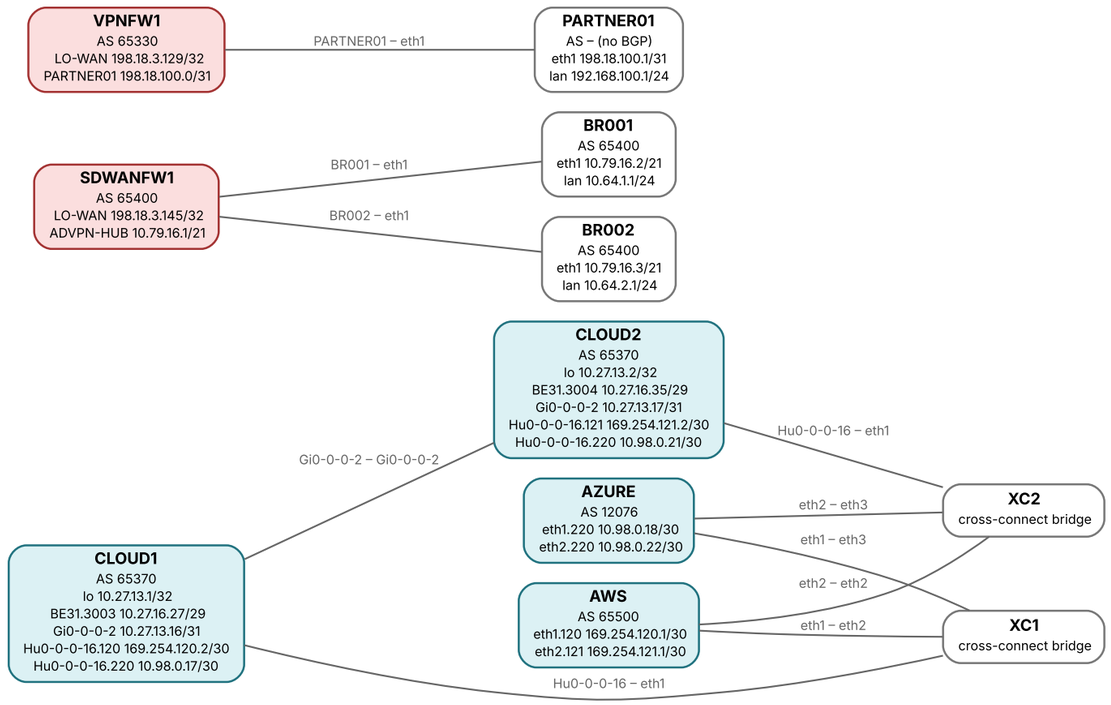
11 · Internet edge, DMZ, clouds and branches

Everything outside the VTEP fabric, with interfaces and addresses from the stand-in scripts

Internet edge and DMZ



Clouds, branches and partner



The edge pair is a layer-2 switch between the firewalls' outside interfaces (VLAN 3200, 10.27.14.0/24) and the ISP routers. ISP1/ISP2 send 0/0 to the Internet FW only while their carrier default exists. Carriers, clouds, branches and the partner are lab-only stand-ins with the public or provider ASNs of the design (192.0.2.0/24 and 198.18.0.0/15 are documentation and benchmark ranges).

12 · VRFs and traffic paths

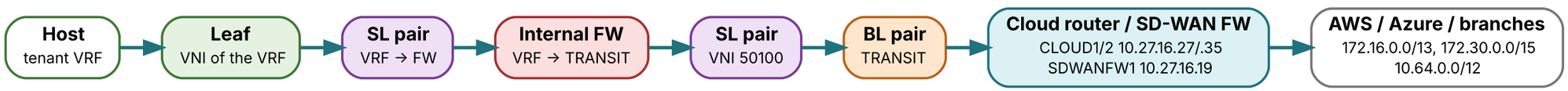
Where each VRF lives and the three ways traffic leaves the fabric

VRF	L3 VNI	Switches	On	DC3 summary sent to the Internal FW
PROD	50001	22	L1–L20, SL1, SL2	10.24.0.0/18
DEV	50002	22	L1–L20, SL1, SL2	10.24.64.0/18
CITRIX	50022	22	L1–L20, SL1, SL2	10.24.144.0/20
PARTNER	50031	4	BL1, BL2, SL1, SL2	10.24.160.0/21
REPLICATION	–	0	no switch yet	
TRANSIT	50100	4	BL1, BL2, SL1, SL2	

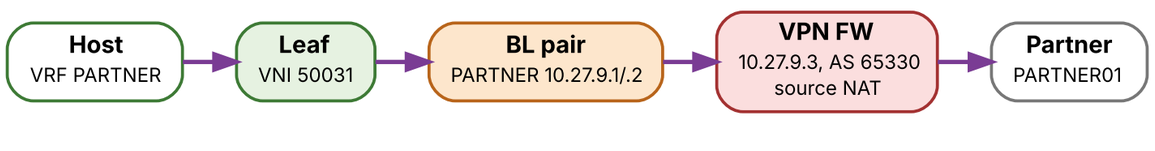
Tenant to the internet (the path the failure tests ping: PROD host to 198.18.1.10)



Tenant to the clouds and branches (never through the Internet FW, failure test 16)



Partner VRF to the VPN partner



Tenant VRFs leave the fabric only through the Internal FW on the service leaves; only TRANSIT reaches the border leaves' outside attachments. REPLICATION is defined (VNI 50041) but no switch carries it yet. The path labels are written from docs/DESIGN.md section 7 and checked against the configs; the VRF table is read from them. Captured packets of path 1: docs/VXLAN-EVPN-CAPTURE.md.

13 · Routing policy

Every route-map on the firewall, cloud and SD-WAN attachments (service pair and border pair; both members are equal)

Pair	VRF	Neighbor	Dir	Route-map	Permits (prefix-list: entries)
SL pair	CITRIX	DC3-CITRIX-INTFW	in	RM-CITRIX-FROM-FW	PL-DEFAULT: 0.0.0.0/0
SL pair	CITRIX	DC3-CITRIX-INTFW	out	RM-CITRIX-TO-FW	PL-CITRIX-SUMMARY: 10.24.144.0/20
SL pair	DEV	DC3-DEV-INTFW	in	RM-DEV-FROM-FW	PL-DEFAULT: 0.0.0.0/0
SL pair	DEV	DC3-DEV-INTFW	out	RM-DEV-TO-FW	PL-DEV-SUMMARY: 10.24.64.0/18
SL pair	PARTNER	DC3-PARTNER-INTFW	in	RM-PARTNER-FROM-FW	PL-DEFAULT: 0.0.0.0/0
SL pair	PARTNER	DC3-PARTNER-INTFW	out	RM-PARTNER-TO-FW	PL-PARTNER-SUMMARY: 10.24.160.0/21
SL pair	PROD	DC3-PROD-INTFW	in	RM-PROD-FROM-FW	PL-DEFAULT: 0.0.0.0/0
SL pair	PROD	DC3-PROD-INTFW	out	RM-PROD-TO-FW	PL-PROD-SUMMARY: 10.24.0.0/18
SL pair	TRANSIT	DC3-TRANSIT-INTFW	in	RM-TRANSIT-FROM-FW	PL-TRANSIT-SUMMARY: 10.24.144.0/20, 10.24.64.0/18, 10.24.160.0/21, 10.24.0.0/18
SL pair	TRANSIT	DC3-TRANSIT-INTFW	out	RM-TRANSIT-TO-FW	PL-DEFAULT: 0.0.0.0/0
BL pair	PARTNER	DC3-PARTNER-VPNFW	in	RM-PARTNER-FROM-FW	PL-PARTNER-NAT: 10.24.160.0/21 ge 28 le 28
BL pair	PARTNER	DC3-PARTNER-VPNFW	out	RM-PARTNER-TO-FW	PL-DEFAULT: 0.0.0.0/0
BL pair	TRANSIT	DC3-TRANSIT-INETFW	in	RM-TRANSIT-FROM-FW	PL-DEFAULT: 0.0.0.0/0
BL pair	TRANSIT	DC3-TRANSIT-INETFW	out	RM-TRANSIT-TO-FW	PL-TRANSIT-SUMMARY: 10.24.144.0/20, 10.24.64.0/18, 10.24.160.0/21, 10.24.0.0/18
BL pair	TRANSIT	DC3-TRANSIT-SDWANFW	in	RM-TRANSIT-FROM-SDWAN	PL-BRANCH-SUMMARY: 10.64.0.0/12
BL pair	TRANSIT	DC3-TRANSIT-SDWANFW	out	RM-TRANSIT-TO-SDWAN	PL-TRANSIT-SDWAN-OUT: 10.24.144.0/20, 10.24.64.0/18, 10.24.0.0/18; PL-CLOUD: 172.16.0.0/13 le 24, 172.30.0.0/15 le 24
BL pair	TRANSIT	DC3-TRANSIT-CLOUD1	in	RM-TRANSIT-FROM-CLOUD	PL-CLOUD: 172.16.0.0/13 le 24, 172.30.0.0/15 le 24
BL pair	TRANSIT	DC3-TRANSIT-CLOUD1	out	RM-TRANSIT-TO-CLOUD	PL-DEFAULT: 0.0.0.0/0; PL-TRANSIT-SUMMARY: 10.24.144.0/20, 10.24.64.0/18, 10.24.160.0/21, 10.24.0.0/18
BL pair	TRANSIT	DC3-TRANSIT-CLOUD2	in	RM-TRANSIT-FROM-CLOUD	PL-CLOUD: 172.16.0.0/13 le 24, 172.30.0.0/15 le 24
BL pair	TRANSIT	DC3-TRANSIT-CLOUD2	out	RM-TRANSIT-TO-CLOUD	PL-DEFAULT: 0.0.0.0/0; PL-TRANSIT-SUMMARY: 10.24.144.0/20, 10.24.64.0/18, 10.24.160.0/21, 10.24.0.0/18

Route-map (all BGP switches)	Permits
RM-CONN-2-BGP (redistribute connected, default VRF)	PL-LOOPBACKS: 10.27.0.0/24 eq 32, 10.27.1.0/24 eq 32, 10.27.2.0/25 eq 32

Read from the configs of SL1 and BL1; SL2 and BL2 carry the same policy. All entries are permit; anything not matched is denied by the implicit deny at the end of each route-map.

14 · Tenant segments

42 server segments on every server leaf (L1..L20), same VLAN, VNI and gateway everywhere

VLAN	Name	L2 VNI	VRF	Anycast gateway
1010	DC3-PROD-1010	11010	PROD	10.24.10.1/24
1011	DC3-PROD-1011	11011	PROD	10.24.11.1/24
1012	DC3-PROD-1012	11012	PROD	10.24.12.1/24
1013	DC3-PROD-1013	11013	PROD	10.24.13.1/24
1014	DC3-PROD-1014	11014	PROD	10.24.14.1/24
1015	DC3-PROD-1015	11015	PROD	10.24.15.1/24
1016	DC3-PROD-1016	11016	PROD	10.24.16.1/24
1017	DC3-PROD-1017	11017	PROD	10.24.17.1/24
1040	DC3-PROD-1040	11040	PROD	10.24.40.1/24
1041	DC3-PROD-1041	11041	PROD	10.24.41.1/24
1042	DC3-PROD-1042	11042	PROD	10.24.42.1/24
1043	DC3-PROD-1043	11043	PROD	10.24.43.1/24
1044	DC3-PROD-1044	11044	PROD	10.24.44.1/24
1045	DC3-PROD-1045	11045	PROD	10.24.45.1/24
1046	DC3-PROD-1046	11046	PROD	10.24.46.1/24
1047	DC3-PROD-1047	11047	PROD	10.24.47.1/24
1048	DC3-PROD-1048	11048	PROD	10.24.48.1/24
1049	DC3-PROD-1049	11049	PROD	10.24.49.1/24
1050	DC3-PROD-1050	11050	PROD	10.24.50.1/24
1051	DC3-PROD-1051	11051	PROD	10.24.51.1/24
1110	DC3-DEV-1110	11110	DEV	10.24.74.1/24
1111	DC3-DEV-1111	11111	DEV	10.24.75.1/24
1112	DC3-DEV-1112	11112	DEV	10.24.76.1/24
1113	DC3-DEV-1113	11113	DEV	10.24.77.1/24
1114	DC3-DEV-1114	11114	DEV	10.24.78.1/24
1115	DC3-DEV-1115	11115	DEV	10.24.79.1/24
1116	DC3-DEV-1116	11116	DEV	10.24.80.1/24
1117	DC3-DEV-1117	11117	DEV	10.24.81.1/24
1118	DC3-DEV-1118	11118	DEV	10.24.82.1/24
1140	DC3-DEV-1140	11140	DEV	10.24.104.1/24
1141	DC3-DEV-1141	11141	DEV	10.24.105.1/24
1142	DC3-DEV-1142	11142	DEV	10.24.106.1/24
1143	DC3-DEV-1143	11143	DEV	10.24.107.1/24
1144	DC3-DEV-1144	11144	DEV	10.24.108.1/24
1145	DC3-DEV-1145	11145	DEV	10.24.109.1/24
1146	DC3-DEV-1146	11146	DEV	10.24.110.1/24
1147	DC3-DEV-1147	11147	DEV	10.24.111.1/24
1148	DC3-DEV-1148	11148	DEV	10.24.112.1/24
1149	DC3-DEV-1149	11149	DEV	10.24.113.1/24
1150	DC3-DEV-1150	11150	DEV	10.24.114.1/24
1500	CITRIX-DC3-1500	11500	CITRIX	10.24.144.1/24
1501	CITRIX-DC3-1501	11501	CITRIX	10.24.145.1/24

From the DC3-L1 config; every server leaf renders the same list. Gateways use ip address virtual with the anycast MAC 0200.0000.00aa; RT 65300:<VNI> (route-target both).

15 · Management and ZTP

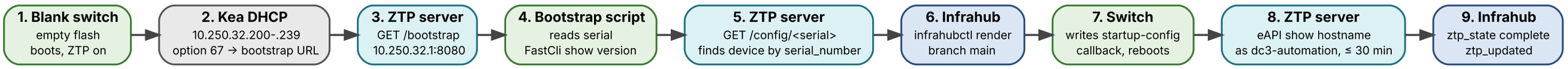
OOB addresses from pool DC3-MGMT0 (10.250.32.0/24, gateway .1) and how a blank switch gets its config

Switch	Management1 (VRF MGMT)	Role
BL1	10.250.32.24	border
BL2	10.250.32.25	border
DMZ1	10.250.32.30	edge/DMZ
DMZ2	10.250.32.31	edge/DMZ
EDGE1	10.250.32.28	edge/DMZ
EDGE2	10.250.32.29	edge/DMZ
L1	10.250.32.4	leaf
L2	10.250.32.5	leaf
L3	10.250.32.6	leaf
L4	10.250.32.7	leaf
L5	10.250.32.8	leaf
L6	10.250.32.9	leaf
L7	10.250.32.10	leaf
L8	10.250.32.11	leaf
L9	10.250.32.12	leaf

Switch	Management1 (VRF MGMT)	Role
L10	10.250.32.13	leaf
L11	10.250.32.14	leaf
L12	10.250.32.15	leaf
L13	10.250.32.16	leaf
L14	10.250.32.17	leaf
L15	10.250.32.18	leaf
L16	10.250.32.19	leaf
L17	10.250.32.20	leaf
L18	10.250.32.21	leaf
L19	10.250.32.22	leaf
L20	10.250.32.23	leaf
S1	10.250.32.2	spine
S2	10.250.32.3	spine
SL1	10.250.32.26	service
SL2	10.250.32.27	service

Management VRF: default route 0.0.0.0/0 via 10.250.32.1, no ip routing vrf MGMT; eAPI (HTTPS) in VRF MGMT; local user dc3-automation.

Zero-touch provisioning (ztp/)

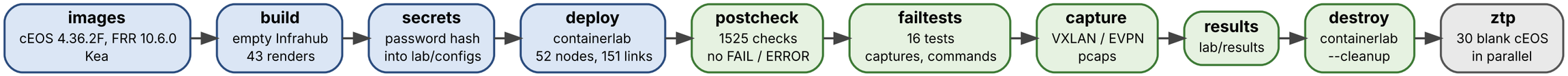


ZTP states: pending (set before racking) → served → complete, or failed. Verified: one switch on the laptop in 66 s; all 30 in parallel, 30/30 complete in 425 s.

16 · Lab and tests

How the lab is run, and what each failure test breaks

The full lab on one host (30 cEOS-lab 4.36.2F, 15 FRR 10.6.0, 5 test hosts, 2 bridges) (docs/QUICKSTART.md, docs/BUILD.md)



#	Failure test	Breaks	Outage limit
1	Carrier loss: ISP-A, then both carriers	ISP-A, ISP-B	5 s, measured
2	ISP router down: DC3-ISP1	ISP1	5 s
3	Spine down: DC3-S1	S1	5 s
4	Border leaf down (DC3-BL1), then service leaf down (DC3-SL1)	BL1, SL1	5 s, 5 s
5	One leaf-spine link down: DC3-L11 Ethernet49/1 → DC3-S1	L11	2 s
6	MLAG peer-link down (DC3-BL1 - DC3-BL2)	BL1	5 s
7	MLAG keepalive down (DC3-BL1 Ethernet24/1)	BL1	0.5 s
8	One FW port-channel member down: DC3-INTFW1 port19	INTFW1	2 s
9	Silent failure toward the Internal FW (netem 100 % loss on port19 and port20)	INTFW1	measured
10	Internal FW stand-in down: DC3-INTFW1	INTFW1	measured
11	Internet FW stand-in down: DC3-INETFW1	INETFW1	measured
12	Filter test: Internal FW announces 10.99.0.0/16 in TRANSIT	INTFW1	0.5 s
13	BGP graceful restart on an ISP router (bgpd restart on DC3-ISP1)	ISP1	0.5 s
14	ISP1 return (DC3-ISP1 down, then back)	ISP1	5 s
15	Node reload with its rendered config: DC3-L11	L11	measured
16	Tenant to cloud and branch via TRANSIT	AWS, AZURE	measured

Full lab (52 nodes): pre-check 1525/1525, 15 of 16 failure tests pass (test 6 open), post-check 0 new failures, ZTP 30/30, rebuild from the docs passes.